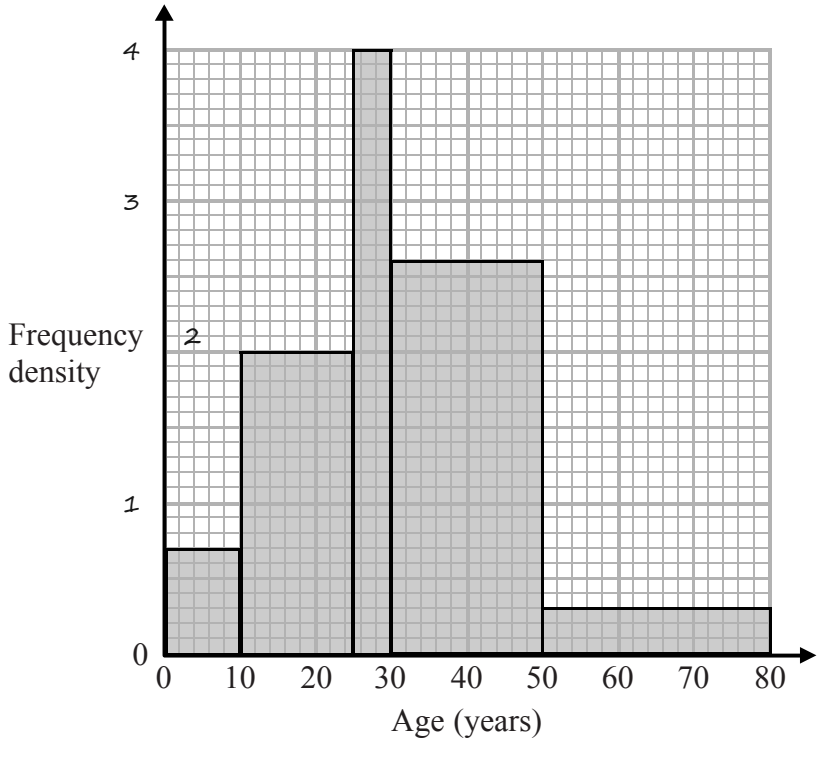
23 The histogram shows information about the ages of the members of a football supporters club.



There are 20 members aged between 25 and 30

One member of the club is chosen at random.

What is the probability that this member is more than 30 years old?

Density for 25–30 = 20 ÷ 5 = 4.

1 mark
Calibrate density

Age	Width	Density	Frequency
0–10	10	0.7	7
10–25	15	2	30
25–30	5	4	20
30–50	20	2.6	52
50–80	30	0.3	9

1 mark
Frequency = width × density

Total = 118; over 30 = 52 + 9 = 61.

$$P(\text{over } 30) = \frac{61}{118}$$

1 mark
61/118

(Total for Question 23 is 3 marks)